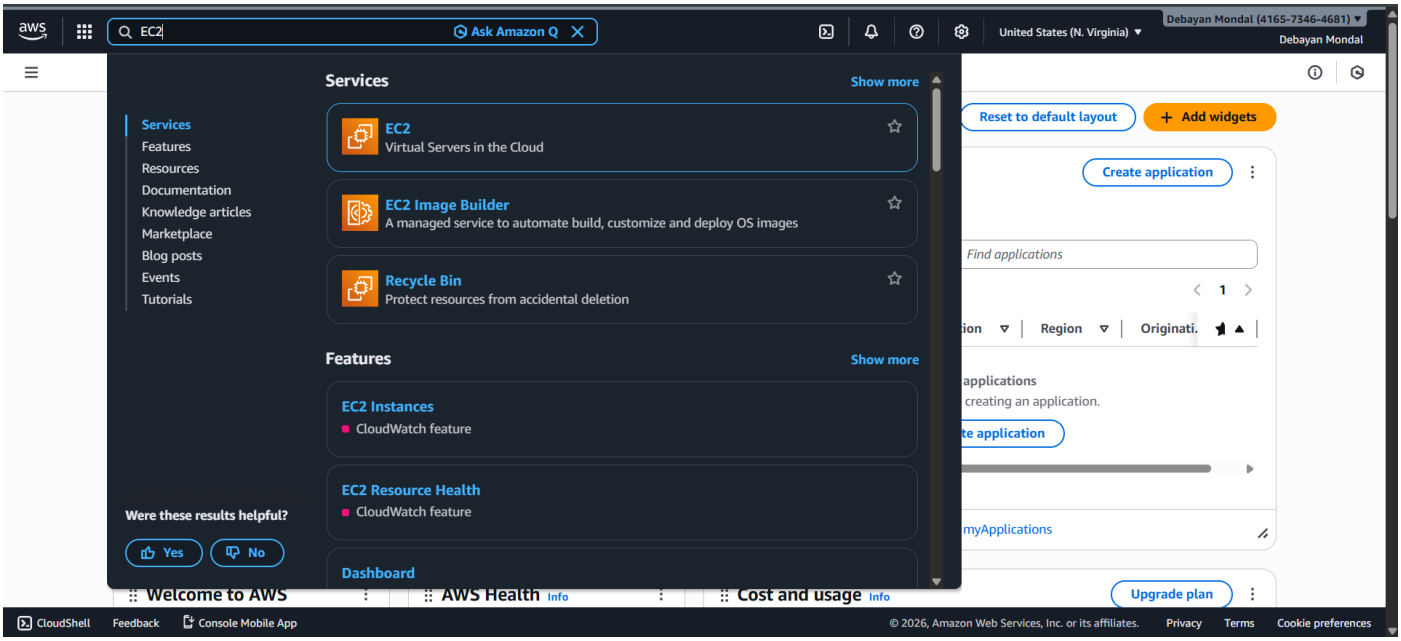
Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Assignment No.:10

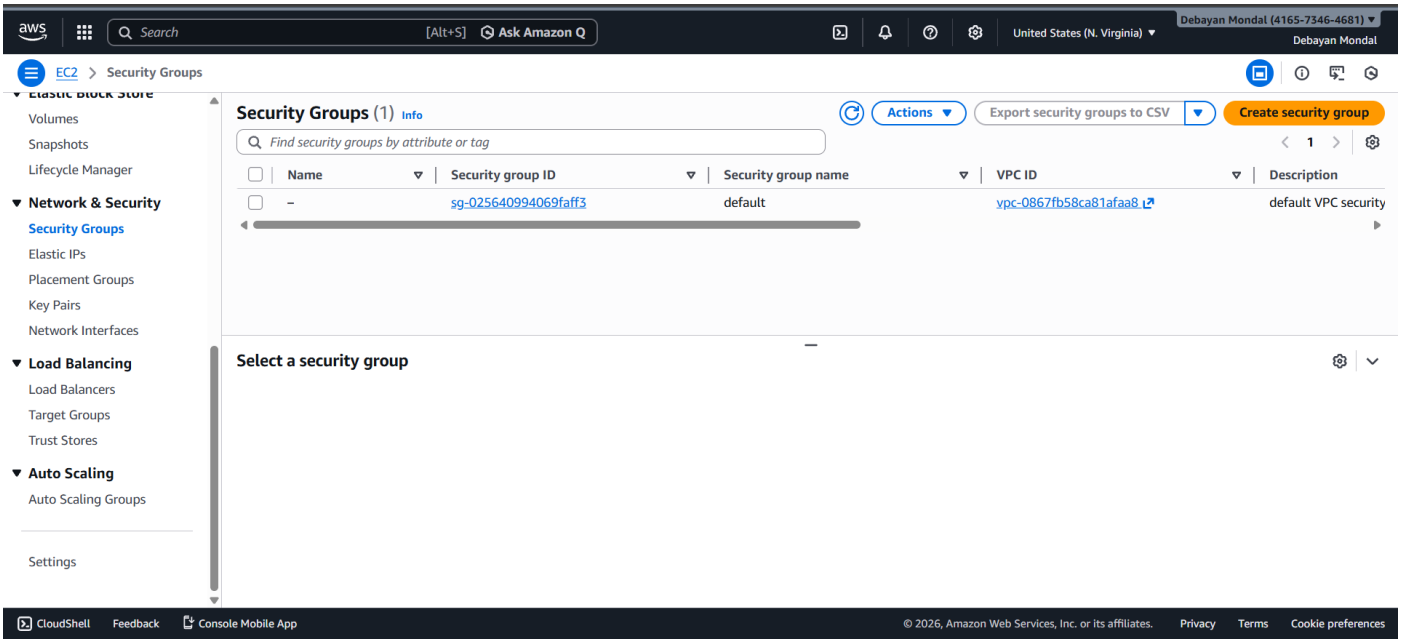
Problem Statement: Deploy a project from Github to EC2 by creating a new security group and user data.

Solution:

Step 1: Search EC2 in the search bar and click on it.

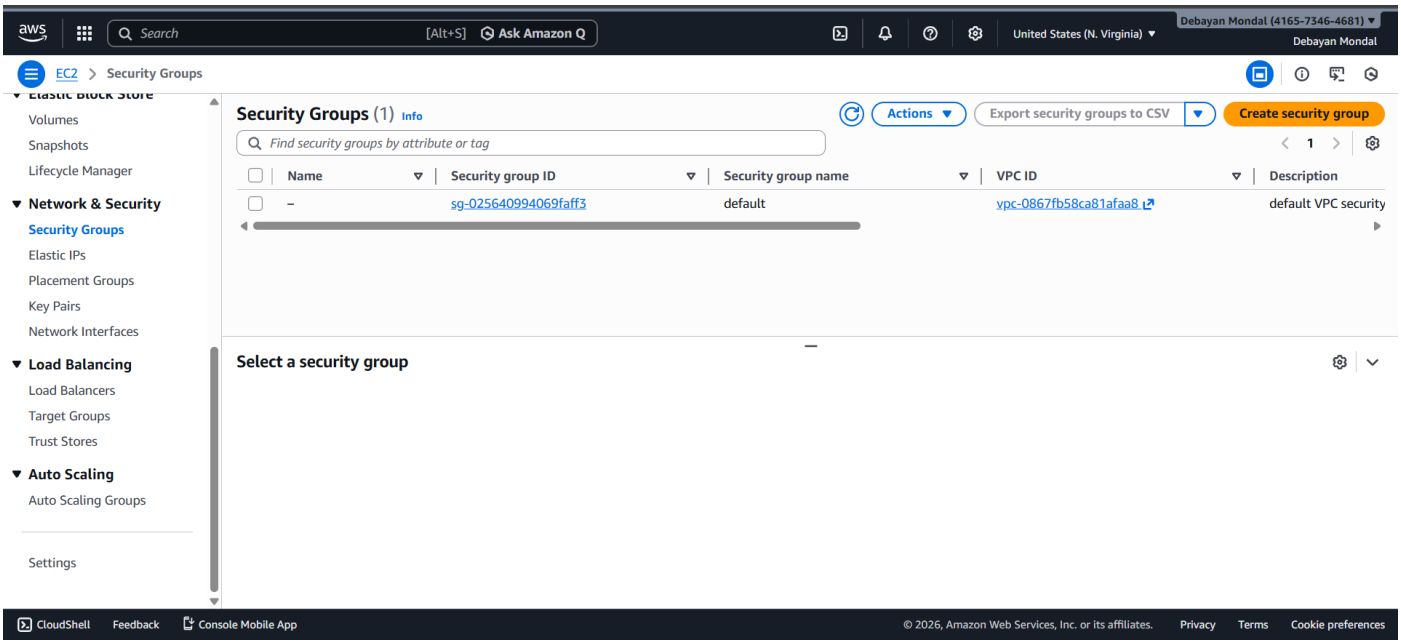


Step 2: In the EC2 page, click on **Security Groups** button from the left panel.

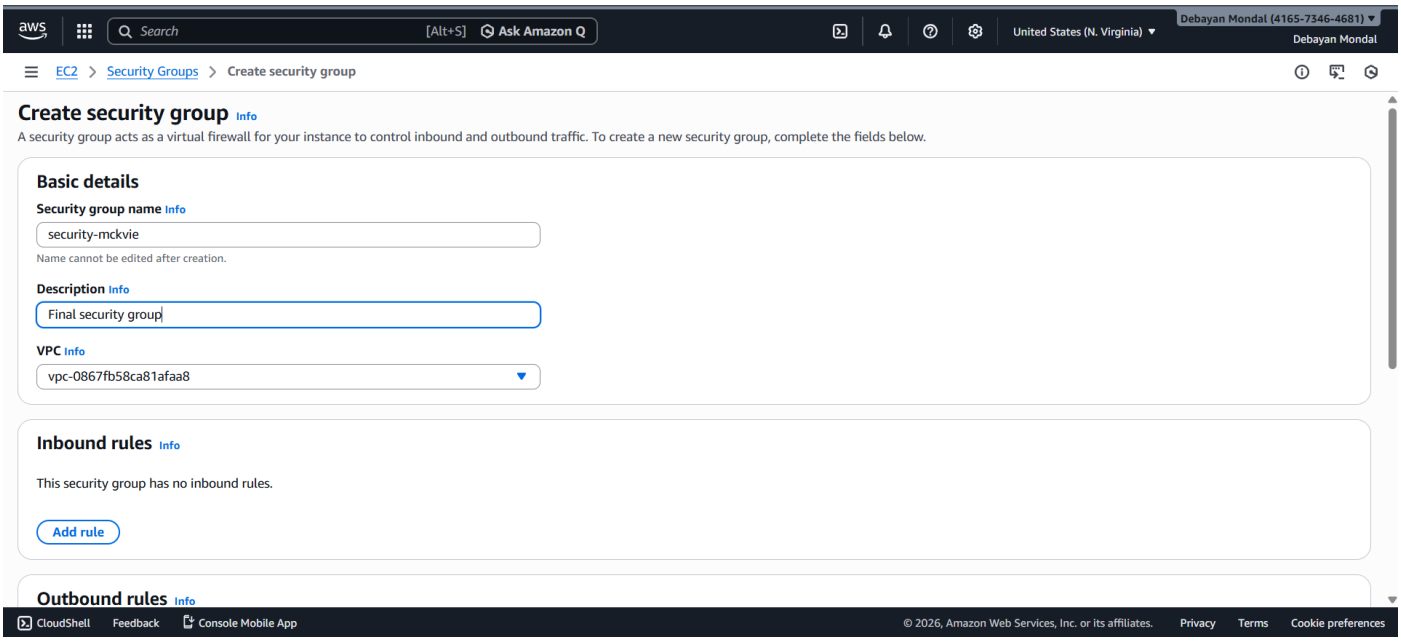


Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 3: Click on the **Create security group** button.



Step 4: Enter the **Security group name**, **Description.** Under the Inbound rules click on the **Add rule** button.



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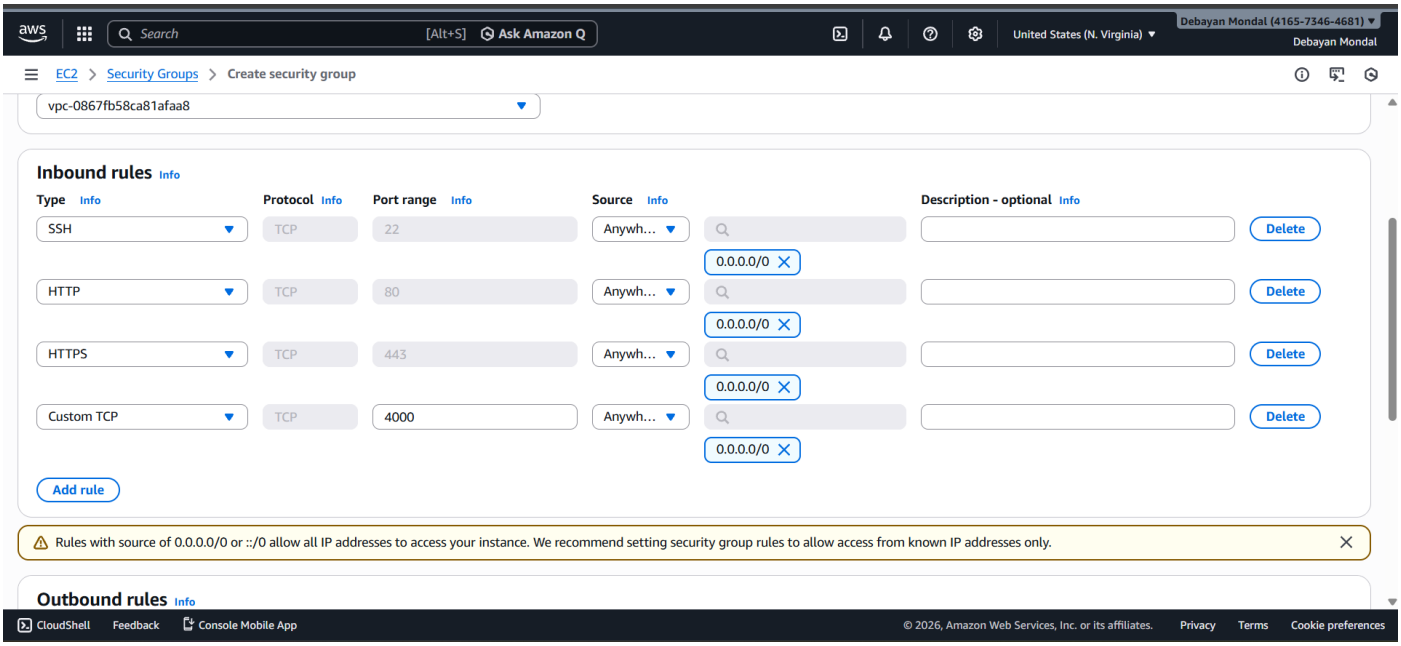
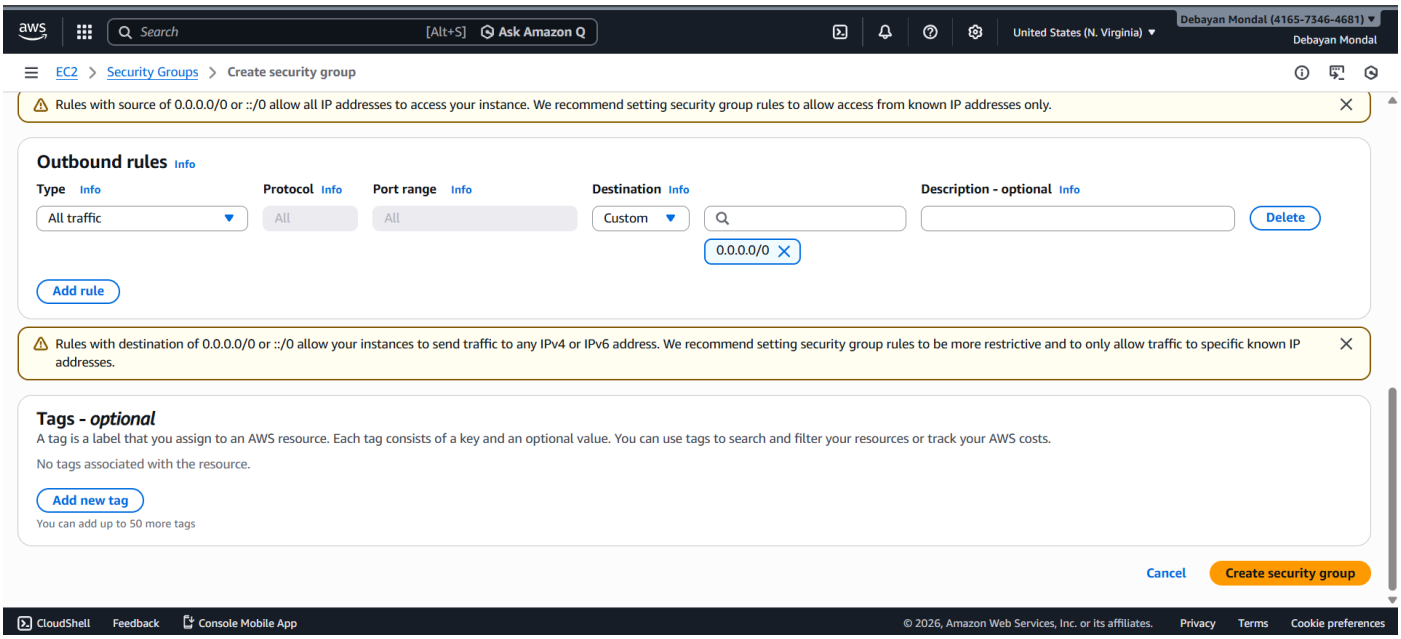
Step 5: Add the types SSH, HTTP, HTTPS, Custom TCP and Enter the port range 22, 80, 443, 4000 respectively.

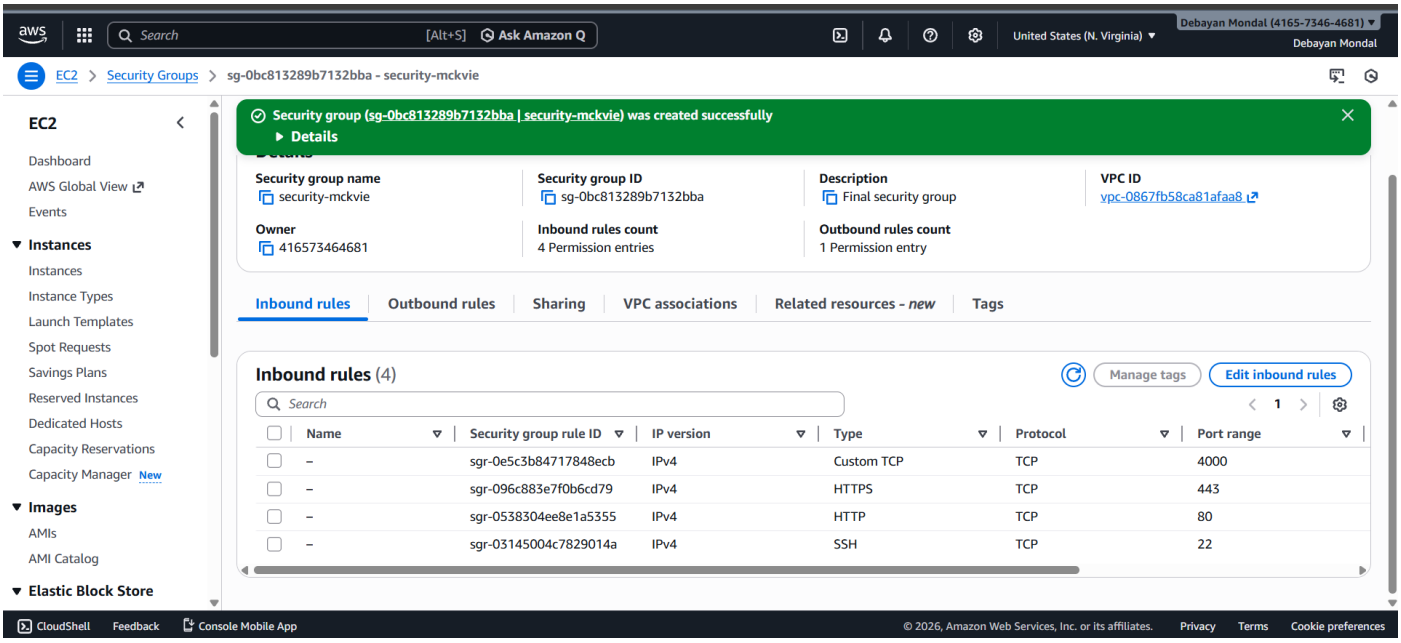


Step 6: Finally click on **Create security group** button.

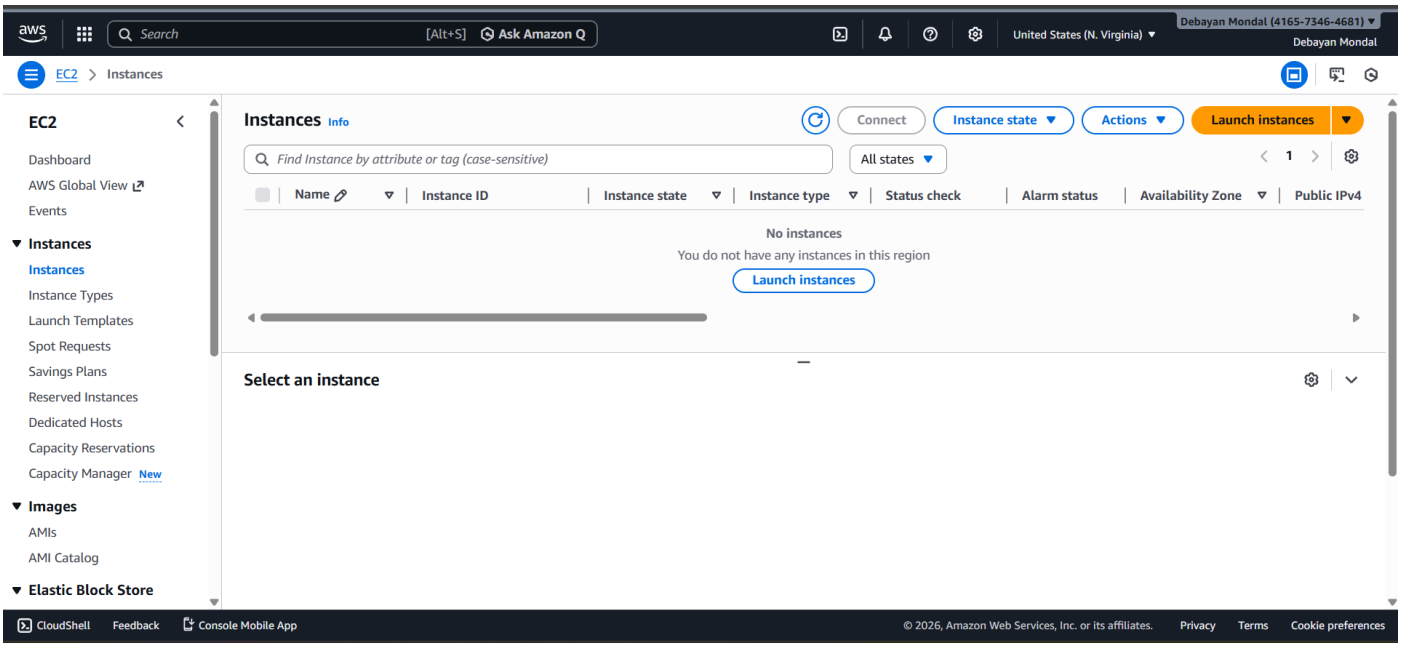


Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 7: The security group will be successfully created. Click on the **Instance** option from the left panel.

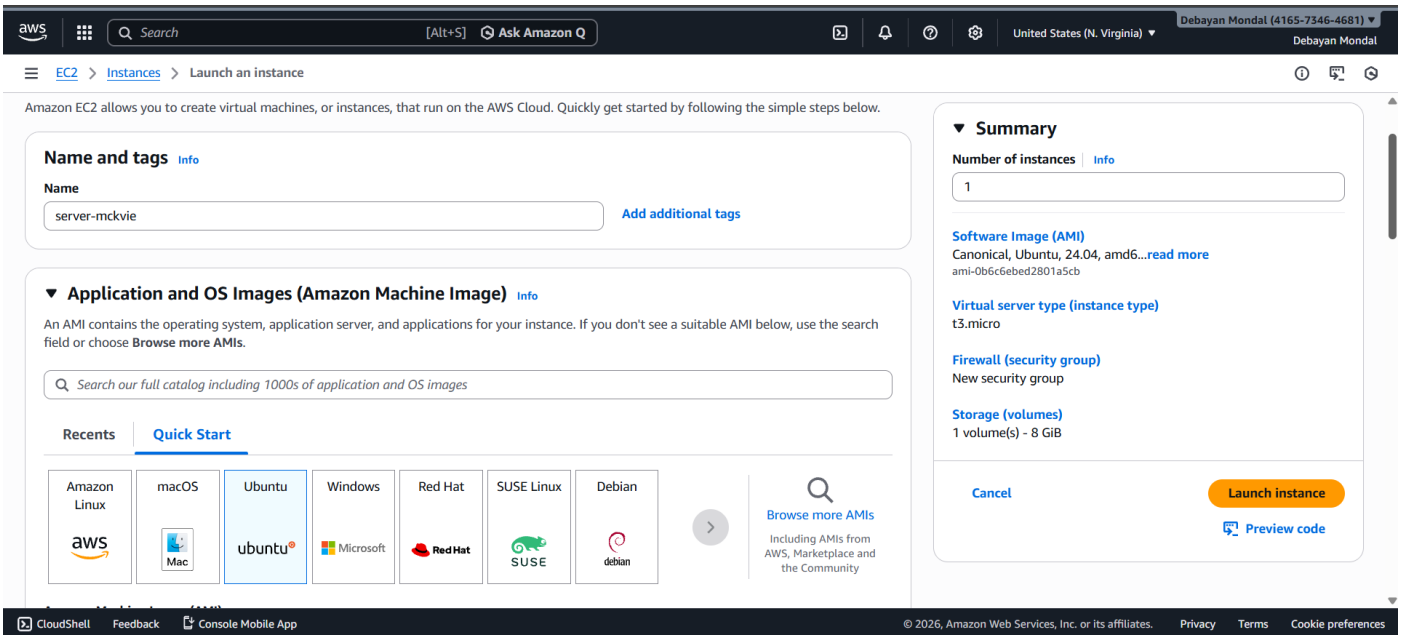


Step 8: Click on the **Launch Instance** button.

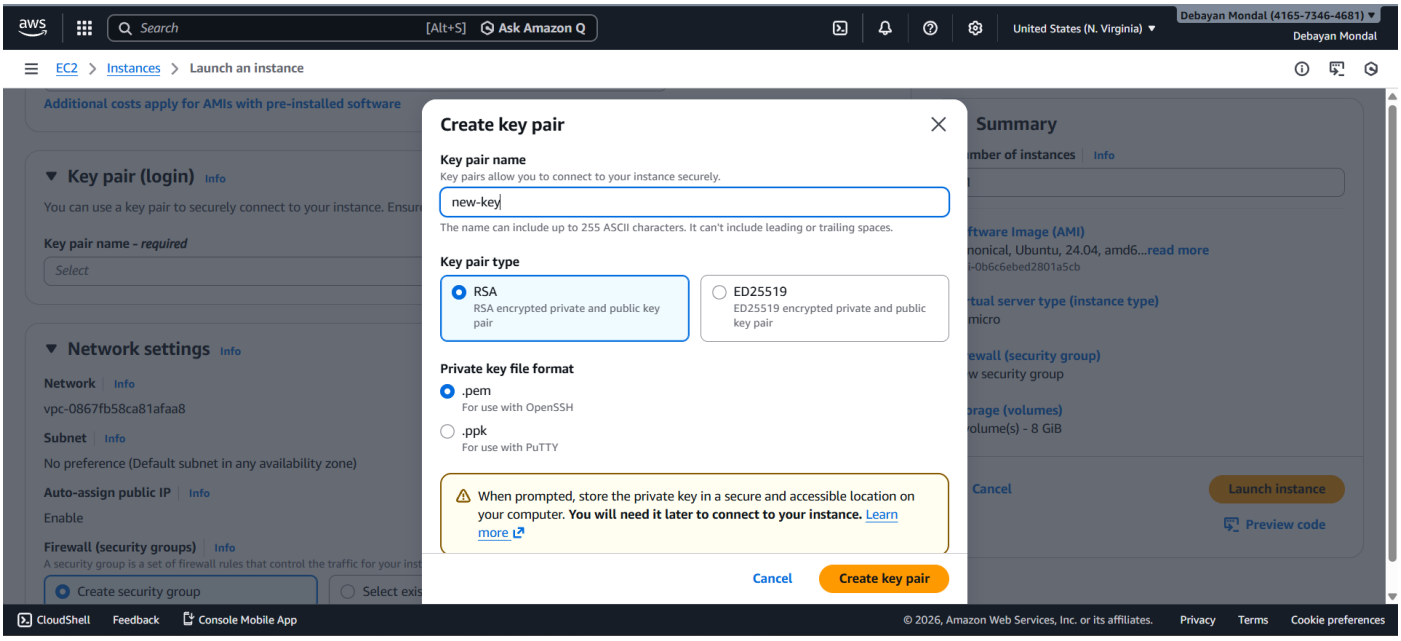


Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 9: Enter the Instance name and Select OS as **ubuntu**.

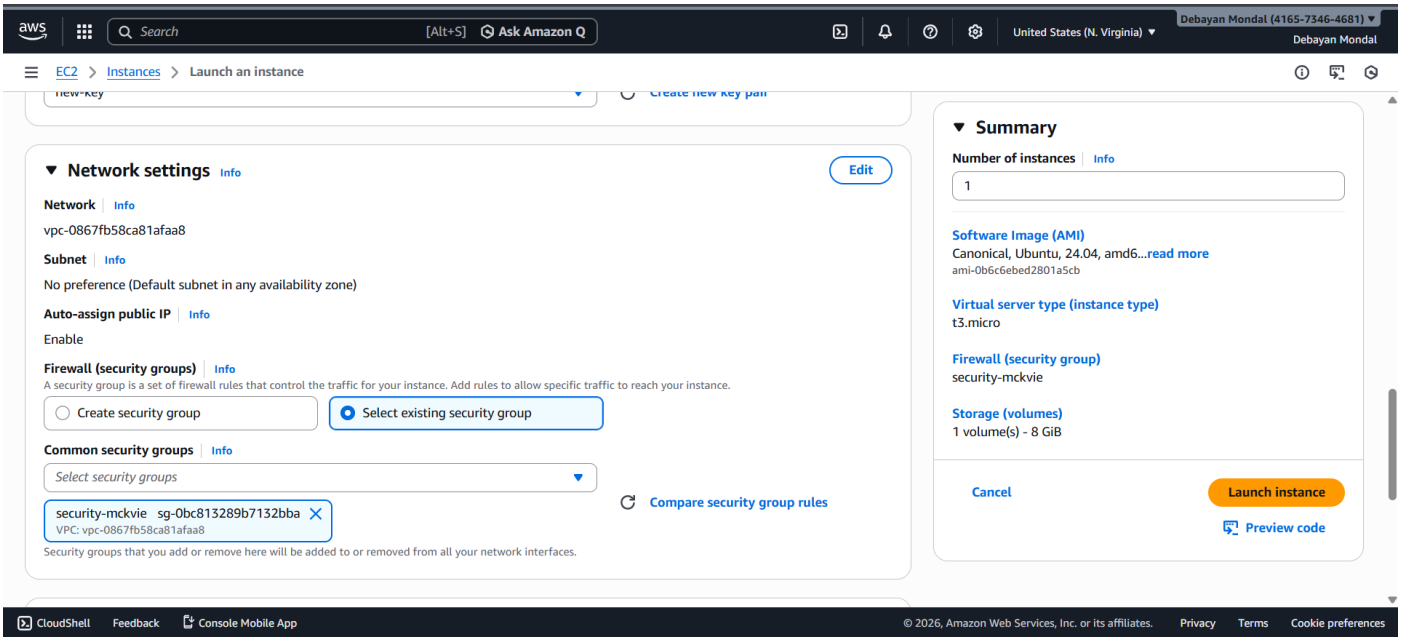


Step 10: Click one create new keypair and then Enter the name of the new keypair then click on the **Create key pair** button. The key will be automatically downloaded into the local device.

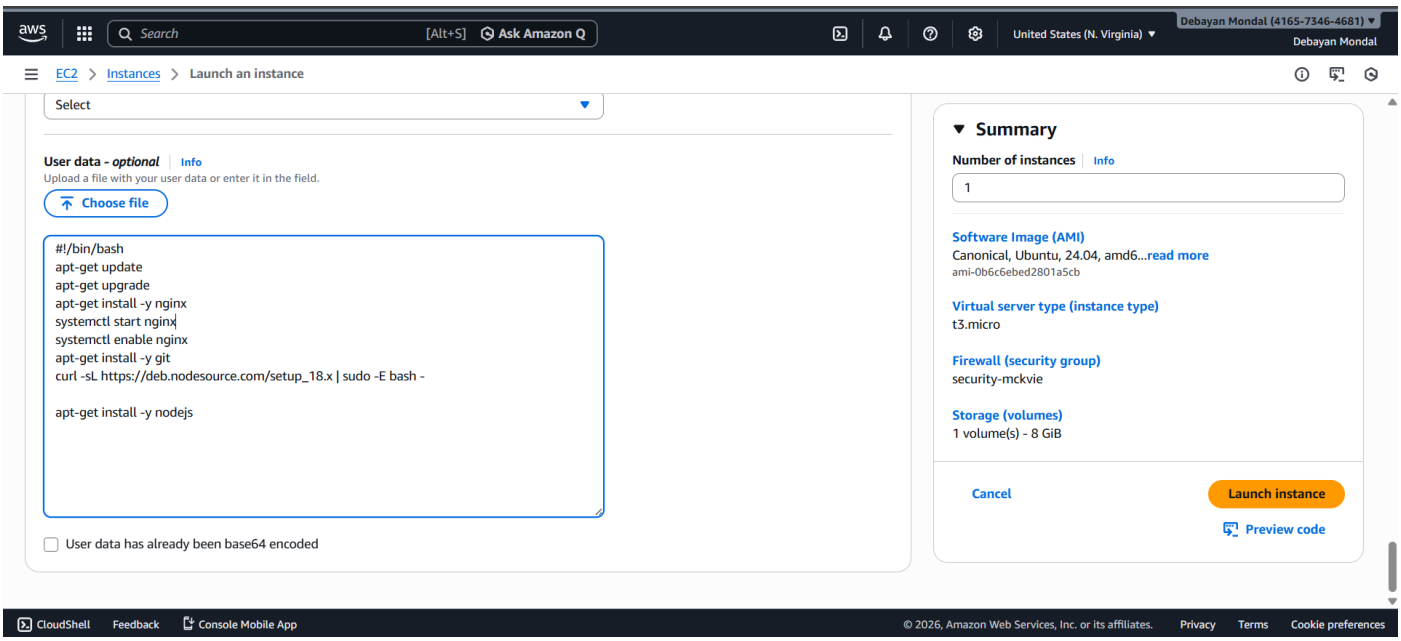


Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 11: Click on **Select existing security group** and then Select the newly created security group from the previous steps.

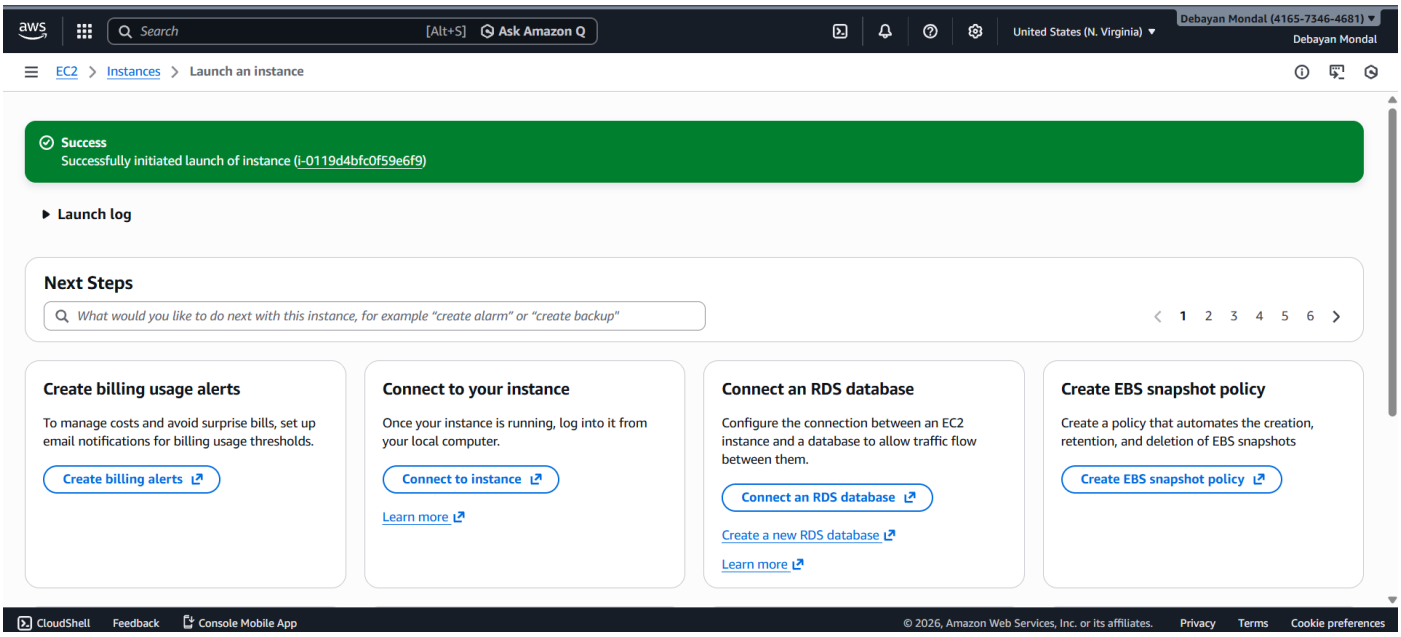


Step 12: Enter the following script inside the User data and then click on **Launch Instance** button.

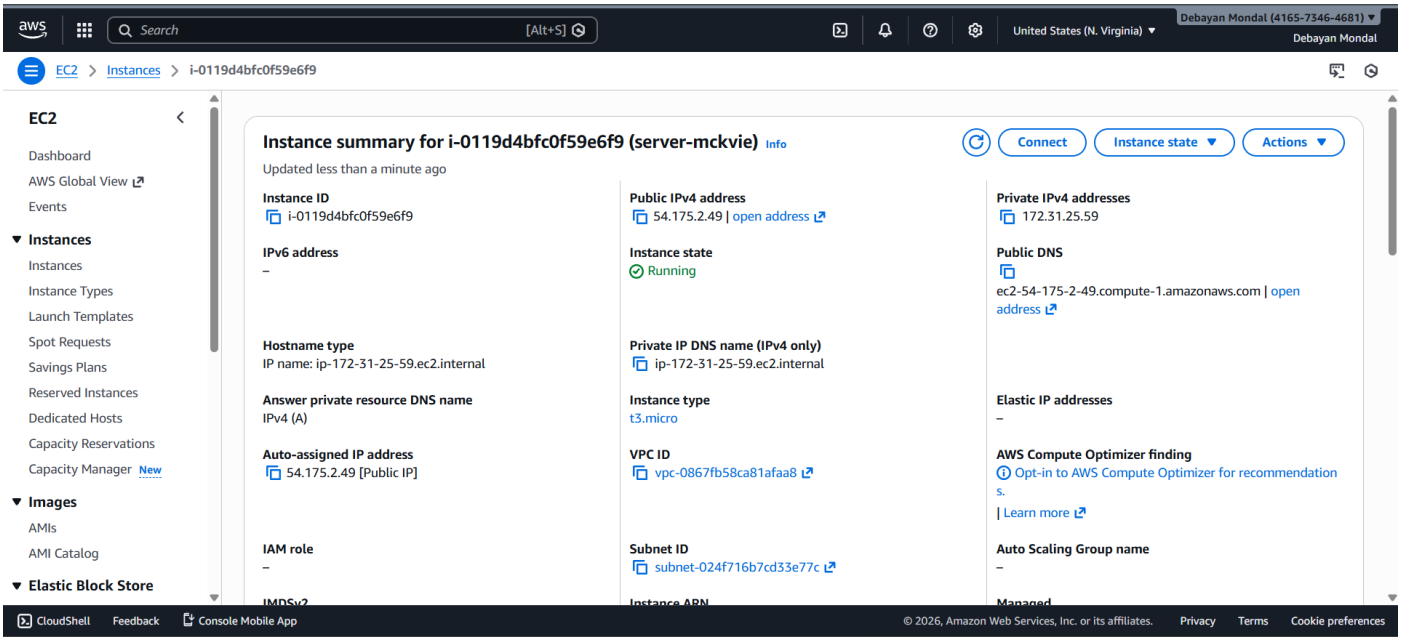


Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 13: The Instance will be successfully created.

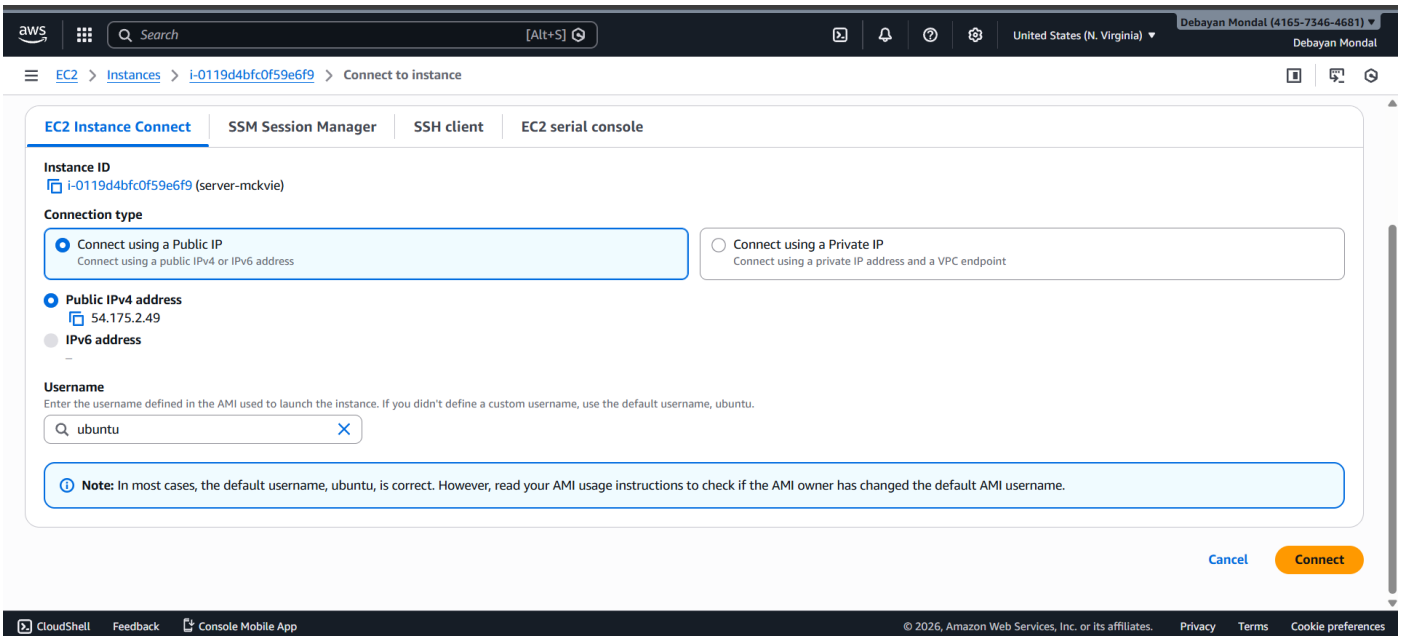


Step 14: Go to the newly created instance and click on the **Connect** button.

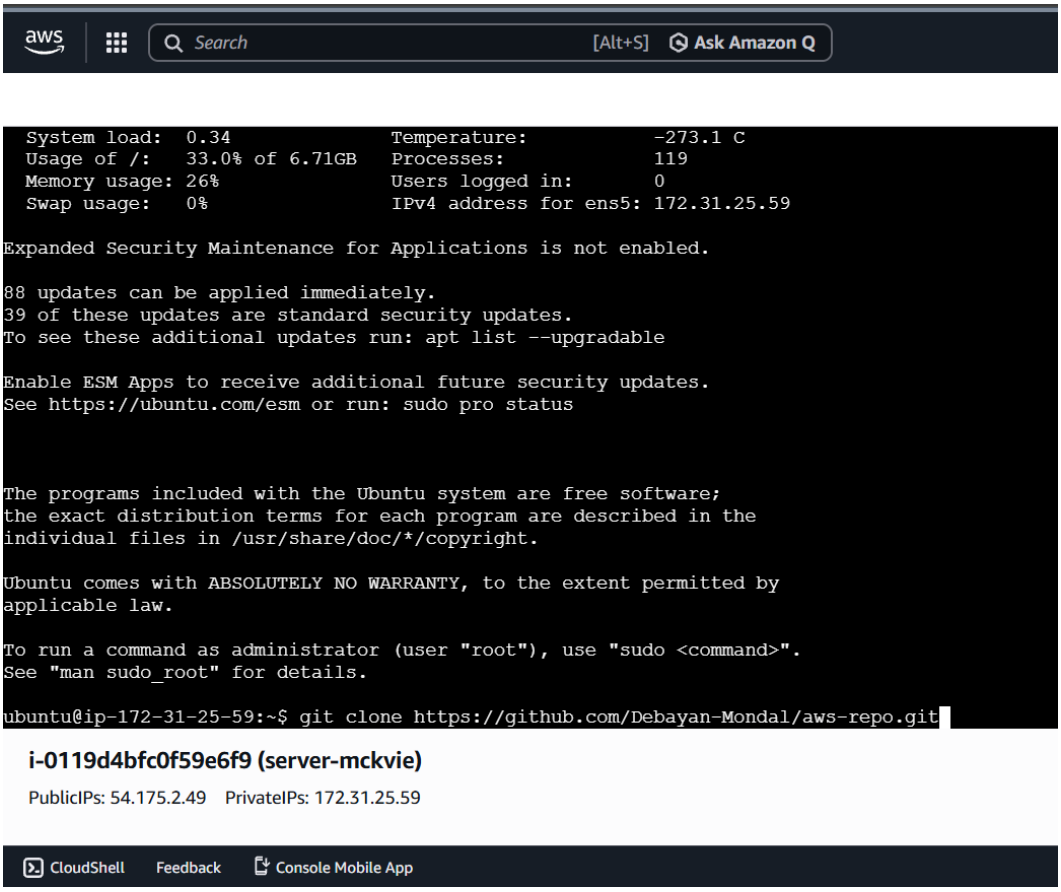


Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 15: Inside this page, click on the **Connect** button.



Step 16: A new tab will be opened with a terminal. Enter the command **git clone https://github.com/Debayan-Mondal/aws-repo.git** to clone the repo containing the project.



Name: Debayan Mondal **Roll. No:** 135 **ExamRoll:** 11600223142

Step 17: Go to the cloned folder and Enter command **ls** to check the files present.

```
remote: Total 14 (delta 4), reused 4 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (14/14), done.
Resolving deltas: 100% (4/4), done.
ubuntu@ip-172-31-25-59:~$ cd aws-repo/
ubuntu@ip-172-31-25-59:~/aws-repo$ ls
index.html  index.js  next.html  package.json  prv.html
ubuntu@ip-172-31-25-59:~/aws-repo$
```

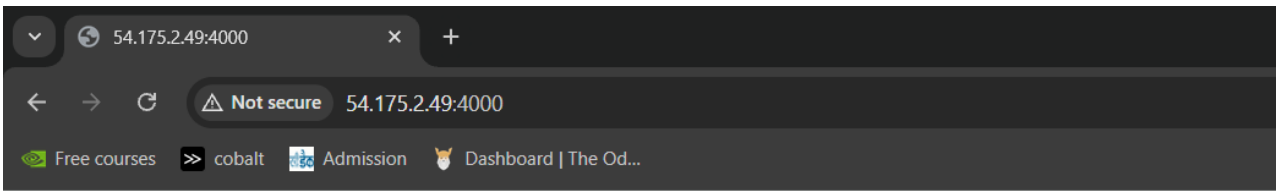
Step 18: Enter the command **npm install** to install the dependencies.

```
ubuntu@ip-172-31-25-59:~/aws-repo$ npm install

added 224 packages, and audited 225 packages in 15s

26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.10.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.10.0
npm notice To update run: npm install -g npm@11.10.0
npm notice
ubuntu@ip-172-31-25-59:~/aws-repo$
```

Step 19: Enter the command **node index.js** to start the server.

```
ubuntu@ip-172-31-25-59:~/aws-repo$ node index.js
Started server
```

Step 20: We can put the public IPv4 adress on the URL address bar a new tab to open the website.



Hello MCKV